Malaysian Genomics Resource Centre

8F Jalan Teknologi, 3/6 Taman Sains Selangor 1,
Kota Damansara (PJU5)
47810 Petaling Jaya, Selangor, Malaysia.

Certificate of Analysis

COA No:

PRODUCT DETAILS

Name	
Batch	
Date	
Cell Source and Passage	Human Umbilical Cord (Wharton's Jelly), Passage 2
Storage Temperature	4 – 8^{o}C
Viable Cell Count	100.7 x 10^{6}

TEST	SPECIFICATION	TEST RESULTS	REMARKS
Infectious Diseases Screening	HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR	Not Detected / Non-Reactive	QUALITY CONTROL APPROVED
Morphology of Cells	Spindle-like cells	Small spindle shaped cells	QUALITY CONTROL APPROVED
Cell Viability	≥90% ± 0.1 of requested cells	Within range	QUALITY CONTROL APPROVED
Sterility	No cloudiness of the test media	Negative	QUALITY CONTROL APPROVED
Bacterial Endotoxin Test (LAL)	Endotoxin sensitivity 0.125 EU/mL	Negative	QUALITY CONTROL APPROVED
Mycoplasma	Mean of absorbance <0.05	Negative	QUALITY CONTROL APPROVED
Immunophenotyping	Positive marker – CD73, CD90, CD105 (≥ 95%) Negative marker – CD11b, CD19, CD34, CD45, HLA-DR (≤ 2%)	Within range	QUALITY CONTROL APPROVED
Differentiation Capacity	Minimum tri-lineage differentiation	Positive stain on differentiated cells	QUALITY CONTROL APPROVED
Karyotyping	Normal cells = 46 chromosomes	Normal	QUALITY CONTROL APPROVED

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Authorised Signature:

Quality Control Department
Date:

Certificate of Analysis

COA No:

Supporting Information

MORPHOLOGY OF CELLS

Figure 1: Mesenchymal stem cells as viewed under an inverted phase contrast microscope (10x).

STERILITY TEST [6, 8]

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.

Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST [3,6]

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

MYCOPLASMA TEST [4,6]

Negative findings for mycoplasma species *M. hyorhinis, M. arginini, M. fermentans, M. orale, M. Pirum, M. hominis, M. salivarium and A. laidlawii.*

IMMUNOPHENOTYPING

Marker	Positive			Negative (CD11b, CD19, CD34, CD45, HLA-DR)
	CD73+	CD90+	CD105+	
Result (%)				

Table 1: Flow cytometric analysis was performed to verify the characteristic surface marker expression profile of mesenchymal stem cells.

DIFFERENTIATION CAPACITY [1]

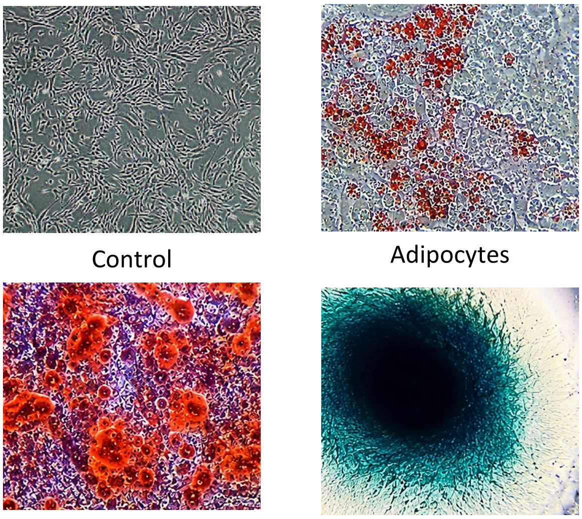

Figure 2: In-vitro differentiated stem cells observed under an inverted phase contrast microscope (10×). The cells demonstrated successful differentiation into adipocytes, osteocytes and chondrocytes.

KARYOTYPING [7]

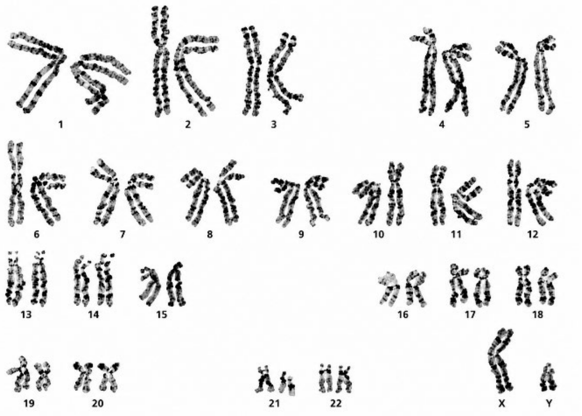

Figure 3: Karyotype analysis of the mesenchymal stem cells demonstrated a normal chromosomal complement of 46, XY. No numerical or structural chromosomal abnormalities were detected. The cells exhibit a normal diploid karyotype.

9001:2015

REFERENCES

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